

Jean Lucien Randrianantenaina

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[in](#) LinkedIn · [GitHub](#) · [Portfolio](#)

Education

- Sep 2025 – Present **PhD in Computer Science**
University of Bath, United Kingdom
- Jan 2024 – Dec 2024 **MSc in Machine Learning and AI**
Stellenbosch University, South Africa
- Sep 2022 – Jun 2023 **MSc in Mathematical Sciences**
African Institute for Mathematical Sciences (AIMS), Cameroon
- 2020 – 2024 **MSc in Mathematics**
University of Fianarantsoa, Madagascar
- 2015 – 2019 **BSc in Mathematics**
University of Fianarantsoa, Madagascar

Experience

- Jan 2025 – Sep 2025 **ML/AI & Backend Developer**
Arato, Fianarantsoa, Madagascar
- Designed and deployed retrieval-augmented generation pipelines and chatbot systems.
 - Built automation workflows with Selenium and n8n for large-scale data collection and operational support.
 - Developed back-end services and REST APIs.
- Jan 2025 – Sep 2025 **Part-time Lecturer**
University of Fianarantsoa, Madagascar
- Delivered teaching in Linear Algebra and Mathematics for AI to undergraduate and postgraduate students.
 - Prepared lecture materials, coursework, and assessments.

Technical Skills

- Programming **Python, JavaScript, SQL, Bash, HTML/CSS**
- Frameworks & Libraries **PyTorch, TensorFlow, JAX, Scikit-Learn, LangChain, HuggingFace, Django, Flask, FastAPI, Node.js, Vue.js, Selenium**
- Database & Tools **MySQL, PostgreSQL, MongoDB, ChromaDB, Git, Docker, ~~LaTeX~~**
- Languages **English (Intermediate), French (Fluent), Malagasy (Native)**

Selected Projects

- Nov 2025 **Gym Management Web Application**
GI-Fitness, Ifarihy, Antananarivo, Madagascar
Built a full-stack gym management web application using Django REST Framework for the backend and React for the frontend.
- Nov 2024 **Conditional Flow Matching in Feature Space PDF** [↗](#)
MSc research project
Investigated simulation-free training of continuous normalising flows in latent feature space under the supervision of Dr Shane Josias and Prof. Willie Brink.
- Jul 2024 **Reinforcement Learning Suite** [GitHub](#) [↗](#)
Implemented Q-Learning, SARSA, and DQN, and benchmarked them on MiniGrid using Gymnasium, PyTorch, and Weights & Biases.

Jul 2024 Neural Machine Translation (English–Afrikaans) GitHub [↗](#)

Fine-tuned Transformer-based models for technical translation, improved tokenisation choices, and evaluated performance with BLEU.

Feb 2024 Recommender System

Developed a probabilistic matrix factorisation model for MovieLens 25M and accelerated training with NumPy and Numba.

Jan 2024 Optimisation of UNet for Brain Tumour Segmentation using Genetic Algorithms

MSc research project

Investigated architecture optimisation for brain tumour segmentation using genetic algorithms under the supervision of Dr Aurelle Tchagna Kuanou and Dr Joelson Randriamparany.

Publications & Talks

- 2023 **There Is No Carmichael Number of the Form $2^n p^2 + 1$**
Florian Luca and Jean Lucien Randrianantenaina, Integers, Volume 23 [↗](#)
- 2023 **Fermat's Last Theorem, with $n = 4$**
Talk, AIMS Cameroon

Awards & Scholarships

- 2025 **University Research Studentship Award (URSA)**
University of Bath
 Full PhD funding awarded through competitive selection.
- 2024 **Google DeepMind Scholarship**
Stellenbosch University
 Competitive scholarship for MSc study in machine learning and AI.
- 2023 **Industry Immersion Programme Scholarship**
AIMS / ESMT Berlin
 Fully funded programme in business management.
- 2022 **Mastercard Foundation Scholarship**
AIMS Cameroon
 Competitive scholarship for MSc study in mathematical sciences.

Certifications

- 2023 **Business Management**
ESMT Berlin – Industry Immersion Programme
 Soft skills and management training in Limbé, Cameroon.
- 2022 **Certified Back-end Developer**
SAYNA & OIF
 Six-month bootcamp covering Node.js, SQL, and modern web technologies.